SAMSUNG

Chapter Quiz

Coding and Programming

Together for Tomorrow!
Enabling People
Education for Future Generations

Quiz

Chapter 3

For Student

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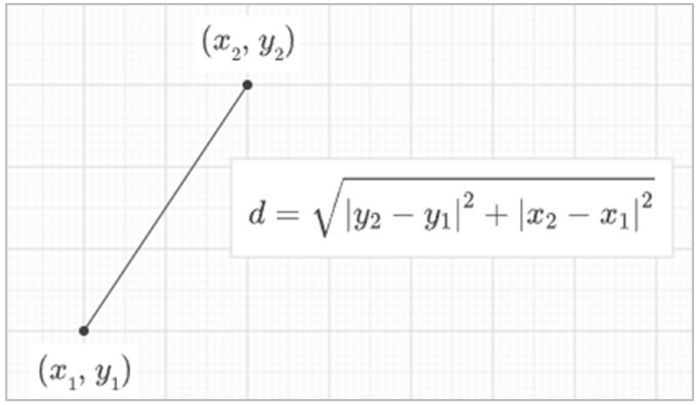
Q. 17-01

Receive the coordinates (x1, y1), (x2, y2) of two points from the user and print the distance between the two points as follows. To do this, implement the distance(x1, y1, x2, y2) function.
Coding guideline: Refer to the figure below for the formula to find the distance between two points.

```
Enter the x1 coordinate: 0
Enter the y1 coordinate: 0
Enter the x2 coordinate: 3
Enter the y2 coordinate: 4
The distance between two points:  5.0
```



Q. 18-01

A palindrome is a sentence, word, or string that is the same as reading any character from back to front. Words like level, eve, and radar are examples of the palindrome.
Let's use a recursive call to determine a palindrome. Define a function called is_palindrome and write a program that receives a string from the user and prints whether the string is a palindrome or not.
Coding guideline: Call the is_palindrome function inside the is_palindrome function.

Q. 19-01

There is a list named n_list with values of [10, 20, 30]. Write a code that puts n_list into the map function and returns the result as follows.
Coding guideline: Use the map and lambda functions.

```
Twice the input value:  [20, 40, 60]
Three times the input value:  [30, 60, 90]
```

Q. 20-01

In the following code, expect what value will finally be included in num_list. Calculate the expected result by hand.
Coding guideline: Through the for statement, the output will be printed 5 times. Expect this output and write the result.

```
def calc_digit(n):
    def final(digit):
        return digit**n
    return final

num_list=[]
for num in range(1,6):
    num_list.append(calc_digit(num))
    print(num_list[num - 1](num))
```

Q. 21-01

Implement multiplication (*) and division (/) of two vectors using the mul and truediv methods learned in unit 21. Assuming that v1 is (30, 40) and v2 is (10, 20), write a code that prints the output as below as the result of the multiplication and division of two vectors.
Coding guideline: Write a code that extends the functionality of the Vector2D class you learned in this chapter.

```
v1 * V2 =  (300,800)
v1 / v2 =  (3.0,2.0)
```